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PERSPECTIVE

The Future Walks With Us: How Robots Could Transform Pedestrian Dynamics and Urban Safety

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ABSTRACT The vision of smart cities is steadily advancing towards realization, with the expectation that this innovative concept will become a widespread reality in the near future. Robotic systems, including service robots, companion robots, and automated vehicles, are essential components of this vision. The integration of robotic systems into our society offers a valuable means of supporting individuals when conventional approaches prove insufficient or completely inadequate. This paper illuminates how robots are currently utilized in pedestrian assistance, specifically in evacuation, flow control, and guidance. The review indicates that robots can positively impact pedestrian dynamics in various ways, including reducing congestion near bottlenecks, guiding route selection, aiding crowd merging, enhancing evacuation efficiency, and providing responsive guidance in unpredictable environments. Meanwhile, significant challenges still exist. These involve developing dependable crowd models, including robot impact, ensuring safe and socially acceptable human-robot interactions, achieving reliable perception in crowded settings, addressing ethical and legal issues, and translating simulation results into real-world applications. According to this paper, effective pedestrian management in the future will likely necessitate integrated systems that leverage robotics, artificial intelligence, sensing, and control. If future research systematically overcomes technical, behavioral, and implementation hurdles, these systems could become valuable tools for both daily crowd operations and emergency responses.

INDEX TERMS Crowd management, intelligent transportation, pedestrian traffic, robot.

I. INTRODUCTION

Since ancient times, people have gathered for ceremonies, rituals, and competitions, culminating in large assemblies. The absence of any considerable challenges in crowd management on those specific days can be attributed to the availability of abundant open space and the relative simplicity of the environment. With the passage of time, many of these gatherings remain visible, even as the world's population grows, physical space becomes more limited, and environmental issues become more complex. Aside from these traditional locations, we now observe overcrowding in places like metro stations, event venues, and shopping malls. Insufficient guidance and regulation can lead to crowd disorder, which may result in blockages and inconsistent pedestrian flow during pedestrian aggregation [1]. Particularly during emergencies, crowd disorder can escalate into crowd accidents, such as stampedes, and such incidents are often linked to pedestrians' competitive behavior. The incident that took place on February 5, 2004, on the Mihong Bridge in China, along with the events that unfolded in Cambodia on November 22, 2010, and India on January 15, 2011, serve as some of the most indicative examples of disasters that have occurred in the last few years, in addition to the stampedes that happened on New Year's Eve of 2013 in Angola and the Ivory Coast [2], [3], [4]. These accidents have resulted in hundreds of deaths and thousands of injuries. Mitigating congestion enables the secure, comfortable, and

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efficient movement of individuals while also contributing to the development of a sustainable society.

The capacity to save lives in emergency situations is the reason why managing pedestrian crowd dynamics has recently drawn a lot of attention. From the perspective of the transportation community, it is generally understood that modifying pedestrian facilities can improve efficiency and overall safety. For example, the inclusion of obstacles can stabilize flow patterns and promote fluidity, while the implementation of zigzag-shaped geometries and columns can mitigate pressure in panicking crowds. Helbing et al. [5] conducted a pedestrian experiment in a panic scenario, and the results demonstrated that the presence of an obstacle adjacent to an exit increased pedestrian flow by 30%. Furthermore, simulations conducted by Escobar and De La Rosa [6] indicate that obstacles positioned near an exit result in, at most, a twofold increase in pedestrian flow. Further investigation into these studies has revealed that the location of an obstacle significantly influences the extent of increased pedestrian flow [6], [7], [8], [9], [10], [11], [12], [13]. The research by Shi et al. [14] provides evidence that the size of an obstacle affects its performance. Furthermore, the configuration of an obstacle is critical to pedestrian egress; cylindrical obstacles enhance pedestrian evacuation more effectively than triangular, quadratic, or diamond prism shapes [15]. It is important to consider the height of an obstacle, because an obstacle that can be easily stepped over will affect the pedestrian flow differently compared to one that necessitates a detour, as noted in [16]. A notable limitation of a fixed obstacle is the potential to reduce visibility at the exit, thereby exacerbating pedestrian panic in an emergency [14]. That said, modifying infrastructure can often be costly, and it is not easily reconfigurable when needed in real time.

In the modern era, technology is consistently transforming our daily experiences as new and advanced tools are introduced, becoming captivating and integral aspects of people's lives. A broad range of civilian and industrial systems frequently employ intelligent control strategies and optimization algorithms. The implementation of numerous ideas is facilitated by artificial intelligence and machine learning [17]. The widespread adoption of autonomous agents in society has been propelled by advancements in robotics and AI over the last several decades. As robots become more common in places where humans live, work, and interact, we have started to see mixed crowds made up of robots and entities that act of their own free will. For example, autonomous and driver-assisted cars regularly share the road with human-operated vehicles, and affordable service robots, such as delivery bots and outdoor cleaners, are present in densely populated pedestrian zones. Smart city projects are increasingly focusing on deploying autonomous robots for various tasks, such as street cleaning and public security, to reduce human workload while enhancing efficiency and service quality [18]. In this regard, one of the new methods proposed to manage pedestrian flow is the use of robots to assist pedestrians, which has gained popularity recently. Drawing upon investigations of fixed obstructions, initial studies have recommended robots as dynamic obstacles designed to alter the pedestrian environment [19], [20], [21]. In addition to managing pedestrian flow, moving obstacles, like fixed ones, also offer the added benefit of greater adaptability and versatility. The motion of obstacles can be altered by differences in the number of people present, and they will move differently depending on the crowd. The simulations by Eldridge and Maciejewski [19] support the idea that moving obstacles can improve pedestrian flow by influencing exit choices. As a consequence of the moving obstacles' behaviors, the narrow exit experiences reduced pedestrian flow and less congestion. Moreover, Yamamoto and Okada [20] introduce a strategy for robot-assisted pedestrian flow control, in which two robots individually guide two streams of pedestrians before they meet at a right angle, aiming to reduce congestion at the merge. Moreover, in another research context, the robot, integrating crowd images, formulated strategies based on reinforcement learning, enabling it to dynamically manage two converging pedestrian flows as they passed through a bottleneck exit, as detailed in [22] and [23].

It is possible that robots could be designed to act as instant first responders in emergency situations, contacting the appropriate authorities and directing individuals to safety. Robot-assisted pedestrian scenarios offer numerous advantages compared to traditional stationary signage guidance methods [24], [25]. Robots' ability to be deployed in advance allows for immediate approachability, and the integration of mapping technology enables robots to identify optimal routes. Furthermore, robots can access real-time information about dynamic environments and make immediate decisions [26]. The use of mobile robots may be beneficial in hazardous situations that preclude human involvement, including emergencies characterized by communication system failures stemming from power outages, as well as in diverse evacuation scenarios that require active crowd management [3]. Nonetheless, simulation precedes practical application within this domain, and an accurate model is imperative to produce results that closely mirror real-world scenarios. Understanding the intricacies of crowd behavior and how robots might influence it is a fundamental requirement for developing a robot-controlled crowd system. Researchers from diverse fields, including social science and engineering, have studied crowd behavior for years, leading to numerous survey papers and books [27], [28], [29], [30]. There are three main classifications of modeling strategies: microscopic, macroscopic, and hybrid approaches that use both. Microscopic models concentrate on the intricacies of individual entities and the localized interactions among pedestrians, such as collision avoidance. The most commonly used models in this category are social force models, which assume that human interactions are mediated by social

forces [31], [32]. Unlike microscopic models, macroscopic models explain the overall crowd motion without going into the specifics of individual persons [33], [34]. In essence, these models are usually comprised of a system of partial differential equations used to determine how the crowd's density and mean flow velocity change over time. These macroscopic crowd models are uniquely applicable to the study of large groups of people and to the analysis of their characteristics as a group. However, it may be the difficulty of incorporating the function of human-robot interaction (HRI) on a macroscopic level that limits their exploration in robot-guided or robot-assisted operations [26].

With today's cutting-edge technologies, we should strive to make the use of robots during emergencies a reality. To cope with challenging environments, a robot-assisted system needs to include temperature protection, waterproofing, and impact resistance [35]. Integrating artificial intelligence into robots could yield promising insights into their surroundings and enable interactions with humans. The system's performance improves by combining intelligence and robotics; this combination has been shown to enhance a robot's ability to sense and react to its environment, boosting its functionality, efficiency, and overall performance. Consequently, this expands the range of robotic operations, as they can recognize specific objects regardless of their spatial location. Furthermore, the integration of monitoring and sensing techniques with appropriate computing technologies will result in an improved automated guidance or control system. The emerging capabilities of humanoid robots to function in ways similar to humans could also bring about a revolution in robot-assisted control, affecting how robots and humans interact with each other in both practical functionality and emotional experience. However, every emerging technology also presents critical challenges and potential failure modes that must be carefully considered. In the context of robot-assisted pedestrian control, system performance is highly dependent on assumptions that may not hold in real-world conditions. Pedestrian compliance with robot guidance is inherently uncertain, particularly in stressful or emergency situations, where trust, perception, and individual behavior can vary significantly. Moreover, the presence of robots may introduce unintended disturbances to crowd dynamics, potentially creating new congestion patterns or amplifying instability if not properly controlled. From a systems perspective, deployment risks must also be acknowledged, including sensing inaccuracies, communication delays, hardware limitations, and environmental uncertainties. In addition, the increasing reliance on connected and intelligent robotic systems raises concerns related to cybersecurity and vulnerability to malicious interference. These factors highlight the necessity of designing robust and fault-tolerant control strategies that can maintain safe and reliable operation under uncertainty and adverse conditions [36], [37]. The purpose of this research paper is to investigate the practicality of incorporating robots into the management of pedestrian traffic, which will be achieved through the review of the most significant publications on the topic, ultimately to put forward several proposals in order to achieve the goal expressed in the title of the paper, which is 'the future walks with us'. This paper promotes integrating robots into hazardous environments to better preserve human lives and reduce the workload on emergency services during critical situations.

This review is based on a structured literature selection process to ensure coverage of the most relevant contributions in the field of robot-assisted pedestrian control. The literature search was conducted using major scientific databases, including IEEE Xplore, Scopus, Web of Science, and Google Scholar. The search focused on publications from 1995 to the present, reflecting the period during which both crowd dynamics modeling and robotic systems have significantly developed. Keywords used in the search included combinations of ''robot-assisted pedestrian,'' ''crowd dynamics,'' ''robot-assisted evacuation,'' ''robot-assisted guidance,'' ''social robot,'' ''human–robot interaction,'' and ''crowd control''. The initial pool of articles was filtered based on relevance to robot-assisted or robot-influenced pedestrian management. Studies focusing solely on traditional pedestrian modeling without robotic interaction were excluded unless they provided essential theoretical background. The rest of the paper is organized as follows:

An overview of the forthcoming integration of robots into our lives is presented in Section II. The third section presents the majority of important published papers on robots used to assist humans in scenarios involving evacuation, flow control, and guidance. To elaborate on the future direction and discuss the current problems, Section IV of this paper presents the designated conclusion and discussion segment.

II. BACKGROUND ISSUES

The United Nations, in its 2018 Revision of the World Urbanization Prospects, forecasts that by 2050, 68 percent of global inhabitants will reside in urban areas, a significant increase from the current 58 percent and the 30 percent recorded in 1950 [38], [39]. The primary impetus behind urbanization is economic, as individuals seek enhanced career prospects and superior living conditions. These escalating rates of urbanization impose considerable strain on urban infrastructure and present difficulties for current governance methodologies, encompassing crowd and traffic control. Specifically, occurrences such as peak travel times and singular events pose considerable difficulties and hazards. The aforementioned indicators underscore the pressing need for innovative strategies to manage pedestrian and vehicular traffic, particularly in light of recent technological advancements in domains such as the Internet of Things (IoT), big data, and robotics. Designing an intelligent closed-loop control scheme that gathers data, analyzes it, and reacts with the appropriate actuator can facilitate the realization of safe and smart cities. The drive for increased efficiency, robust safety measures, and environmental sustainability in

transportation is spurring the development of novel mobility solutions through emerging technologies, among which robot-assisted mobility systems hold significant promise. The integration of robotic systems into transportation systems, advanced communication networks, and other vehicular technologies offers a viable approach to mitigating sustainability concerns while simultaneously improving the overall transportation experience [40]. The ongoing progress in robotics is facilitating the integration of sophisticated, intelligent machines into existing transportation infrastructure, thereby empowering them to assume responsibilities that have historically been performed by human operators. The application of technologies such as artificial intelligence, machine learning, sensor fusion, and robot-assisted mobility systems could fundamentally change industries like autonomous vehicles, local delivery services, and public transportation, driving remarkable enhancements in efficiency, sustainability, safety, and accessibility far beyond what was previously thought possible. Implementing robotic aids in public transportation, such as buses, trains, and stations, could improve the overall passenger journey [40], [41], [42].

Robots in autonomous vehicles are an exciting breakthrough in transportation, offering the incredible potential to dramatically change how we travel. This technology could truly transform the landscape of autonomous transportation, making it more efficient and accessible for everyone. The incorporation of robotic systems into these vehicles facilitates environmental perception, informed decision-making, and real-time action execution. A paramount determinant of success lies in the advancement of real-tiperception systems, such as LiDAR, radar, and camera technologies, which empower vehicles to acquire immediate environmental data, including the identification and monitoring of other vehicles, pedestrians, and potential hazards. Sophisticated algorithms enable accurate comprehension of the vehicle's environment and operational status by fusing sensor data, a critical component for secure navigation. Moreover, advances in decision-making algorithms and control systems enable self-driving cars to process sensor data, plan journeys, and make appropriate driving decisions. At the same time, advanced vehicle communication systems, such as vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I), enable autonomous vehicles to work together seamlessly [43]. By sharing information with each other and the surrounding infrastructure, these systems help make our transportation network safer and more reliable. Furthermore, the implementation of autonomous vehicles offers an opportunity to enhance traffic flow, alleviate congestion, and improve fuel economy through refined driving patterns and coordinated traffic operations [44], [45], [46], [47], [48].

Through their integration into public transportation systems, robots are poised to significantly enhance the passenger experience by increasing accessibility and improving operational efficiency, unlocking the potential for optimized resource allocation and more effective traffic management, which in turn will contribute to reduced congestion and an overall improvement in system performance. The domain of passenger assistance sees robots playing a crucial role in augmenting numerous operations, such as handling ticketing procedures, responding to frequently posed questions, aiding in wayfinding and route determination, facilitating the discovery of facilities within transportation hubs, and dispensing real-time travel data [40]. The robots are designed with advanced natural language processing abilities and engaging interactive interfaces, which collectively empower them to interact with passengers in a way that is both exceptionally user-friendly and profoundly intuitive. This contributes to an improved passenger experience by providing individualized support, mitigating wait times, and disseminating precise, prompt information. Robots also have the potential to increase accessibility and help people with disabilities or mobility challenges. They can provide customized assistance, such as aiding with boarding and exiting, offering priority seating, and delivering audiovisual notifications. Robots can contribute to public transportation not only through passenger assistance but also by maintaining and monitoring the necessary infrastructure. Robots designed for inspection, outfitted with sophisticated sensors and high-resolution cameras, can independently evaluate the structural integrity of essential public transportation components, including railway tracks, roadways, pedestrian walkways, and other vital infrastructure. In the event of an emergency, these robots can provide crucial assistance by transmitting live video streams to human operators, enabling them to make well-informed decisions and organize efficient responses [42], [49].

More recently, several cutting-edge technological instruments have been adopted to organize pedestrian dynamics, representing a significant development in this area. In scenarios such as navigation, dynamic signage systems offer a distinct advantage by reducing the time users need to make decisions, and when these systems are adaptive and utilize distributed computing, they can significantly improve the speed and safety of evacuation guidance compared to their non-adaptive counterparts [50]. Nonetheless, limitations in real-time perceptual capabilities and dynamic adjustment have led scholars to extend the application of robotic systems to this domain as well [51]. The integration of intelligent sensors with real-time environmental data empowers robots to make informed decisions for pedestrian navigation, enabling them to compute optimal pathways and expedite guiding people to the nearest exits or safe zones during emergencies [52]. In addition to their other functions, robots are equipped to perceive their environment and make immediate behavioral adjustments to uphold crowd safety standards. To illustrate, Van Den Berg et al. proposed a structured approach to establishing adequate criteria for collision avoidance in multi-robot motion planning, drawing on the concept of velocity obstacles [53]. The research by Zheng et al. presented a system designed to

dynamically modify robot conduct to ascertain navigational safety [54]. A human-robot collaborative navigation planner was conceptualized by Khambhaita and Alami, who predicted pedestrians' future movements to inform the robot's trajectory planning and thereby prevent collisions with non-human dynamic obstacles [55]. Concurrently, an examination of robot navigation necessitates a thorough consideration of the potential influence robots may exert on crowd dynamics. To understand the effect of noise on the performance of human groups, Shirado and Christakis incorporated autonomous software agents (robots) with unpredictable behavior into a networked color-coordination game [56]. The research by Riether et al. delved into whether social robots would alter human performance, concluding that the robots might affect performance similarly to how humans do [57]. Also, the study by Zhang et al. examined how the incorporation of humanoid robots in pedestrian environments affects crowd movement [58]. Experiments revealed that humanoid robots can effectively blend into crowds by exhibiting random behaviors, thereby affecting pedestrian paths and interaction dynamics. The experiment focused on how robot crossing behaviors affected pedestrian groups and individuals across various situations. The findings indicated that pedestrians exhibit distinct behavioral tendencies when encountering humanoid robots, compared with their interactions with fellow pedestrians or with inactive objects. Enhancing social adaptability in robots is a key goal when studying their navigation performance, and this is achieved by implementing social perception. This enables robots to exhibit social awareness and adjust their navigation strategies in response to the social environment [59], [60], [61], [62]. Following this, we will delve into the application of robot-assisted pedestrian control across three significant scenarios: emergency evacuations, pedestrian flow management, and directional guidance, illustrating how the incorporation of robots influences pedestrian traffic patterns.

III. ROBOT-ASSISTED PEDESTRIAN CONTROL APPLICATIONS

A. ROBOT-ASSISTED CROWD EVACUATION

Addressing crowd evacuation is a key scientific and societal concern, intrinsically linked to safeguarding lives and averting casualties during emergencies. Researchers have devoted significant effort to examining crowd evacuation at both the tactical level, which encompasses decisions regarding route and exit selection, and the operational level, where behaviors such as herding are observed, as evidenced by multiple scholarly attempts [63], [64]. Controlled experiments and model-based simulations have led to the discovery and subsequent reappearance of self-organization phenomena, including "clogging" and "faster-is-slower" [65], [66], [67]. Extensive experiments and simulations were conducted in order to thoroughly analyze the critical factors that significantly impact the efficiency of crowd evacuation, investigating the influence of model and scenario variables such as the dimensions and number of available exits [68], the spatial arrangement of the facility [12], the level of panic exhibited by individuals [61], and the visibility conditions for pedestrians [69]. Also, significant research has been conducted on the development of effective evacuation strategies and the determination of optimal plans for emergency scenarios across diverse locations, including enclosed spaces [70], airport terminals [71], subway stations [72], and public gathering areas [73]. These positive results and successes achieved here contribute significantly to ensuring public safety and mitigating risks associated with blockages and trampling [1].

In this context, the positive impact of emergency evacuation guidance has been highlighted by various research projects. For instance, signs are commonly placed on large buildings to serve as evacuation guidance, offering both emergency evacuation information and directional assistance for regular navigation [74], [75]. Additionally, individuals who have undergone training and are thoroughly familiar with the building's architectural design and the designated escape paths can provide vital guidance to evacuees, thereby facilitating their safe departure from perilous locations [76], [77]. The implementation of well-structured evacuation guidance systems is widely accepted as a practical approach to boosting the efficiency of emergency evacuations, thereby significantly reducing the incidence of casualties or even eradicating them in most circumstances. Therefore, integrating all functionalities into a cohesive unit enables the development of robotic systems designed to evacuate people. During an evacuation, the robotic system can display signage, direct individuals to safety, and assist individuals with disabilities. In challenging evacuation situations where human deployment is not feasible, mobile robots offer a valuable solution. Their utility extends to emergency situations that disrupt communication networks due to power outages or cyber threats, as well as to contexts that require sophisticated management of crowd movement.

It has been demonstrated that auditory cues are highly effective for evacuees, leading to the development of directional audio beacons specifically designed for this purpose [78], [79]. The sounds emitted by these beacons fall within a frequency range that makes them readily detectable and locatable by our ears. Appropriate beacon placement is crucial for guiding evacuations effectively. This issue has led to one of the pioneering works in the field of robot-assisted evacuation schemes. Shell and Matarić thoughtfully explored how to autonomously deploy a network of these audio beacons using a team of robots [80]. This study outlined a robust distributed deployment algorithm capable of handling single-robot faults and communication failures among the robot team. By employing an uncomplicated model designed to assess the impact of auditory cues on the process of making navigational decisions, they were able to establish that the estimated duration for egress is positively influenced by the

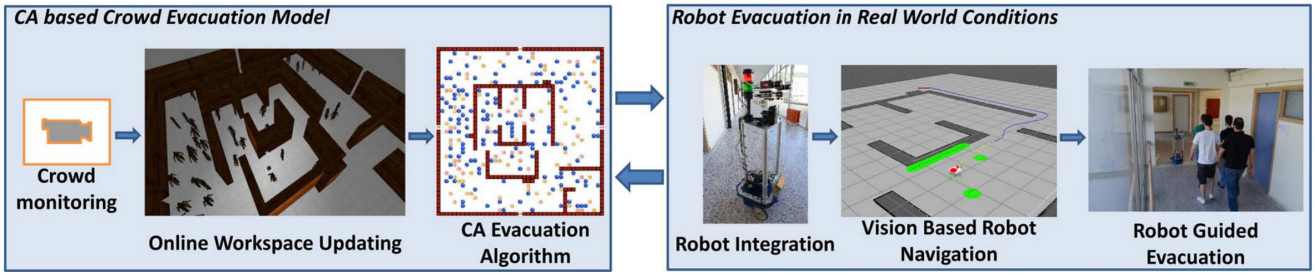


FIGURE 1. The proposed robot-assisted evacuation scheme in [3].

installation of beacons, and furthermore, they discovered that even a limited deployment of these beacons can significantly enhance the efficiency of an evacuation. Subsequent to this research, Boukas et al. presented more sophisticated robot-assisted crowd evacuation protocols for the egress of individuals from indoor to outdoor spaces [3]. Figure 1 illustrates the overall framework of the proposed strategy in [3]. This paper suggested employing a mobile robot to capture the attention of evacuees approaching congested exits and direct them toward less congested ones, thereby facilitating quicker and safer clearance of the area and mitigating the bottleneck effect commonly observed at exit points. The outlined methodology for robot-guided evacuation involves a series of algorithmic steps, beginning with Cellular Automata (CA) based evacuation modeling and human detection via computer vision, followed by path planning. The process involves creating a virtual space for the evacuation zone offline, defining walkable regions, and specifying the layout of obstacles. Thereafter, surveillance cameras engage in online human detection to facilitate the integration of evacuees into the workspace. The simulation provides crucial data on evacuation progression, including the ratio of evacuees per exit at each time interval and the time required for exit clearance. According to the simulation's results, the system calculates a route to redirect evacuees from the most congested exit to an alternative with less crowding, facilitated by the mobile robotic guide. The robot's designated route serves two core purposes: to intercept humans moving towards a congested exit and gain their focus, and to subsequently direct them to a less populated exit [3]. Researchers assessed the proposed robot-guided evacuation model in real-world scenarios, observing notable performance improvements during the critical first-response timeframe. In complex environments, it is crucial to develop an optimal algorithm for robot-assisted emergency evacuations. This algorithm should enable the robot to autonomously identify an optimal escape route to the exit, ensuring the shortest possible evacuation time for those being assisted. Developing a proficient robot-assisted evacuation system requires understanding crowd dynamics in emergency situations. Regarding this matter, the investigation by Tang et al. established that evacuee panic was the principal driver of crowd behavior during emergencies, and this level of panic was subsequently incorporated into their model of evacuee behavior [24]. In emergencies, a little panic can motivate people to move faster, potentially accelerating evacuations. A high level of panic, however, can cause crowd chaos, especially near exits, drastically slowing evacuation. Consequently, the panic exhibited by evacuees reflects their intrinsic motivation to flee, which impacts their interactions with neighbors due to environmental constraints. Quantifying panic by perceived magnitude, this paper proposes a model of panic propagation through a crowd during an evacuation, and then presents a human mobility model incorporating the panic propagation model. Following the model's presentation, an exit-selection algorithm has been devised to identify the exit with the shortest escape duration.

The rush to escape dangerous sites after accidents can, unfortunately, result in chaotic behavior and stampedes. When a crowd is more chaotic, evacuations typically take longer. This implies that insights into chaos can lead to innovative strategies for robots assisting in crowd evacuations. Robots can identify individuals prone to inducing chaotic movements in crowds using chaos modeling to intervene accordingly, such as issuing warnings. Thus, the study in [81] focused on the issue of precision intervention, specifically the optimal timing and location for robot deployment under chaotic conditions. Nevertheless, accurately modeling the evolution of chaos poses challenges, as individuals often move randomly during crowd evacuation. Furthermore, the introduction of robots into crowd environments increases the complexity of crowd dynamics through their regulatory influence. This study approached the solution to this problem by proposing a stochastic optimal intervention method intended for application in robot-assisted crowd evacuation scenarios. The framework's primary structure is organized into three sections: defining and calculating chaos, modeling its stochastic evolution, and addressing stochastic optimal intervention and crowd motion systems. The general scheme of this research is presented in Fig. 2.

The utilization of robotic assistance for evacuations in unforeseen incidents has been the subject of research. Robot-assisted pedestrian evacuation in fire scenarios based on deep reinforcement learning has been studied in [82]. The paper presents an enhanced social force model for pedestrian movement in fire situations and proposes a strategy for

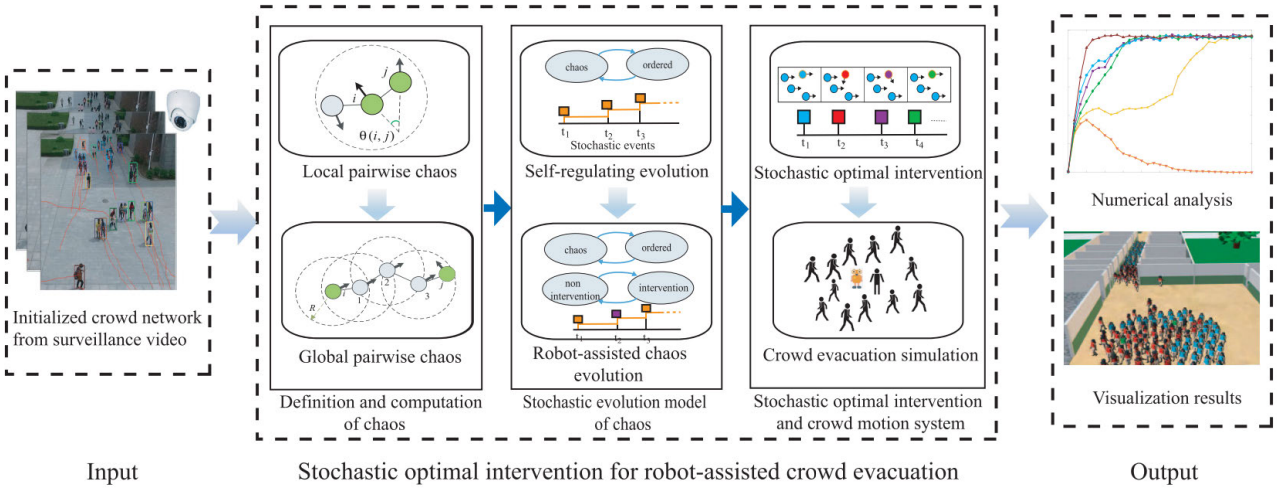


FIGURE 2. The structure of the stochastic optimal intervention method for robot-assisted crowd evacuation [81].

evacuating people using a dynamic robot that employs Deep Reinforcement Learning (DRL). The data demonstrated that when robots are not present, pedestrians navigate around fluctuating fire danger zones and emergency statuses, which precipitate bottlenecks at multiple sites. The DRL-control robot is designed to always find its way to the correct evacuation position, no matter where the fire is, how serious the emergency, or the robot's starting point. Moreover, their research indicates that carefully placed robots can boost evacuation speed by reducing congestion and, in most situations, creating greater separation between people and fire risks, thereby improving pedestrian safety. In scenarios where a large number of individuals need to be evacuated from a specific location, deploying a robot group is essential to facilitate and expedite the evacuation. Therefore, a greater number of robots in the crowd directly results in a more intricate robot control problem. An inconsistency between two robots may result in irreparable consequences for the people. Consequently, the multirobot-guided crowd evacuation has emerged as a subject of significant importance and considerable interest among researchers [83]. One of the initial studies in this area, described in [83], introduced a distributed multi-robot system designed to guide individuals through emergency evacuations. This study's contribution is the presentation of two algorithms designed for evacuation robot team coordination, applicable to both single and multiple teams, and both of which incorporate models of human panic behavior. In the single evacuation robot team, one robot is assigned as the leader role, and the other robots are the shepherds. The responsibility for seeking out evacuees and escorting them to the exit rests on both the leader and the shepherds, who work cooperatively. When multiple teams are involved, the individual leading each team has the authority to interact with the leaders of the other teams to ensure destination information is consistently updated in real time [83]. The study detailed in [26] investigates the challenge of robot-guided crowd evacuation, employing a limited number of robots to direct a substantial human gathering towards secure areas. The fundamental difficulty resides in understanding the methods by which microlevel human–robot interactions can be strategically employed to indirectly shape the behavior of a population that vastly exceeds the number of robots, with the ultimate aim of successfully achieving the collective evacuation objective. To tackle this issue, a dual-scale modeling approach was employed, investigating hydrodynamic models comprising microscopic social force models that delineate localized human movement influenced by individuals, surroundings, and robots, alongside macroscopic equations governing crowd density and flow velocity over time and space. Controllers were developed for the robots, facilitating autonomous environmental exploration for optimal coverage and dynamic adjustments to their local navigation force fields based on real-time crowd macro states to direct individuals to safety [26]. Also, the evacuation of crowds in a railway hub station during emergencies, guided by a group of robots, was investigated in [52]. A cooperative mechanism among robots is incorporated into a hybrid bi-level optimization approach to determine the exit assignments and evacuation paths. Subsequently, a modified social force model is used to construct a dynamical model for robot-guided crowd evacuation. This model incorporates a navigation force to alter the speed and direction of individuals evacuating. Given the continuous increase in urban populations and the growing complexity of city road networks, crowd evacuation and rescue are now paramount concerns for urban areas facing emergencies or unexpected events. The research in [84] investigated large-scale, complex evacuations in urban environments involving a group of robots. An autonomous system-guided approach to crowd evacuation has been

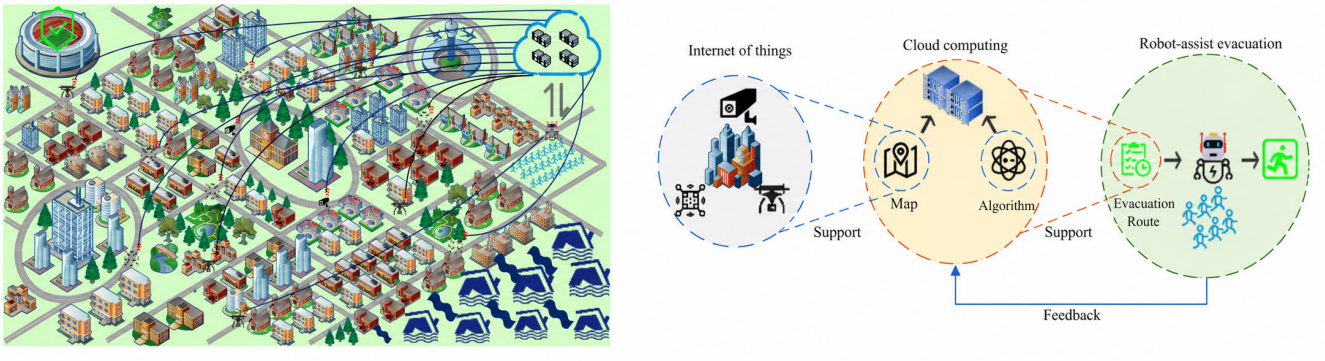


FIGURE 3. The robot-assisted system framework for urban crowd evacuation [84].

presented in [84] to fulfill the stated objective. The presented method equips the robots with the ability to gain a timely and accurate understanding of its environment, thereby enabling the generation of an evacuation map by utilizing cutting-edge information technologies, including the Internet of Things or urban brain. Following this, a more advanced elliptic tangent graph methodology, incorporating both global and local data, was employed to determine a viable and efficient evacuation route for extensive, large-scale environments. A newly developed crowd evacuation model, inspired by the social force model, has been proposed to simulate the real-world process of crowd evacuation in intricate, large-scale environments. To rigorously assess the performance of the proposed algorithm, three real evacuation cases comprising 324 pedestrians have been simulated. The simulation results indicate that an autonomous system's approach to crowd evacuation can identify a more efficient route, thereby reducing the overall evacuation time. Figure 3 illustrates the overall structure of this plan.

B. ROBOT-ASSISTED PEDESTRIAN FLOW CONTROL

Public gatherings involving large populations carry the inherent danger of accidents due to overcrowding in such busy environments. Regulating pedestrian flow is essential at crowded transportation hubs and walkways to mitigate congestion and improve efficiency. Feedback-based control design was presented, utilizing the partial differential equation (PDE) formulation for pedestrian models [85]. In the previous section, we discussed robot-assisted evacuation scenarios. The prevailing approaches in the literature treat a robot as an active directional agent, identifying effective evacuation corridors for people, and assume that a certain number of pedestrians will unconditionally comply with the robot's directives. A new type of robot guide has recently been developed that employs passive HRI. In this scenario, pedestrians perceive the robot as a movable obstruction and adjust their movements accordingly to prevent collisions. The social robot's behavior was optimized using a genetic algorithm to enhance pedestrian flows, as detailed in reference [19]. In the studies presented in [86], [87], and [88], dynamic interaction between a robot and self-organized pedestrian flows was leveraged to effectively regulate the collective swarm behavior of pedestrians. Nonetheless, the aforementioned tasks represent open-loop control, in which the robot's trajectory remains unaltered after planning. Furthermore, these studies employed pedestrian motion models and used offline simulation results to develop robot motion control. Thus, implementing a feedback control system that uses real-time pedestrian movement data and responds to emergent conditions represents a crucial advancement in this area. The research in [17] examined a situation involving a corridor with several entry points, for which a robot is intended to manage the pedestrian flow (see Fig. 4).

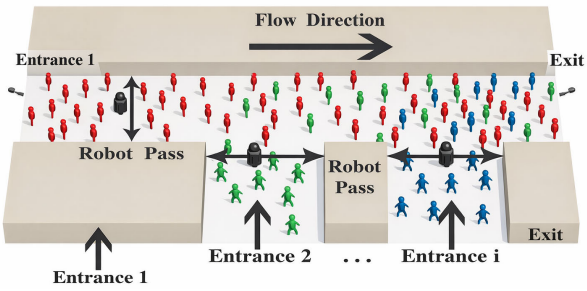


FIGURE 4. The pedestrian flow control scenario defined in [17].

Based on image data acquired from the experimental corridor, the regulated pedestrian velocity was established, and a functional relationship was constructed to correlate the robot's adjustable motion frequency with this velocity. For the modified unidirectional pedestrian corridor with several controlled entrances, they established a macroscopic pedestrian model to achieve optimal traffic flow in the designed scenario. The frequency of the robot's side-to-side motion regulates how many pedestrians enter through each access point. The state feedback controller is then developed, and its stability is assessed using Lyapunov stability theory. The suggested technique facilitated subsequent advances in feedback control in this area. The intricate and unpredictable

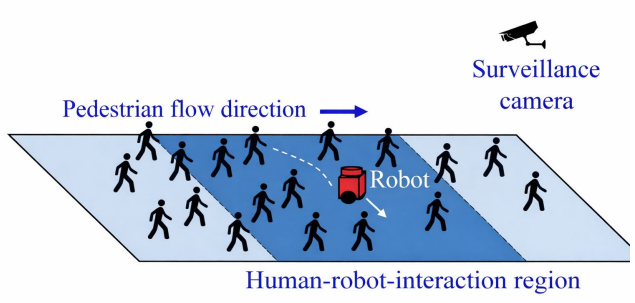


FIGURE 5. The pedestrian flow control scenario defined in [89].

nature of human movement presents significant challenges in accurately modeling pedestrian collective dynamics with current methodologies. This issue was addressed by Jiang et al. in [89] through the implementation of data-driven techniques. In light of recent advancements in camera technologies that enable real-time pedestrian flow measurement, the researchers proposed utilizing this data for feedback control and online learning of HRI to optimize flow regulation. The HRI force was modeled and integrated into the current social force model for pedestrian dynamics. An optimal flow control problem is then formulated to determine the robot's optimal control parameter, regulating pedestrian velocity through dynamic interaction. To address the identified issue, they proposed a control algorithm that leverages learning and adaptive dynamic programming. The system utilized two neural networks: one for robot control (action network) and another to gauge the collective pedestrian flow's performance (critic network). The action network's decision is exclusively determined by the observed data. The scenario considered in this study is depicted in Fig. 5. However, the scenario considered in [89] was not very complicated. Consequently, the researchers proceeded with their investigation and subsequently published a more detailed account in [90]. While prior studies have concentrated on unidirectional corridors, the work presented in [90] delves into the complex interaction of pedestrian flows merging from two perpendicular directions as they pass through a bottleneck (see Fig. 6). The aim was to manage the movement of people entering the area to ensure a smooth and efficient exit through the narrow passage. Furthermore, this study considered crowd pressure [91], a metric that indicates the critical crowd state that could lead to accidents. Analogous to the research in [89], the robot employs learning-based feedback control and optimizes its motion parameters in real time. Unlike the basic unidirectional corridor described in their previous paper [89], the issue here is more complex, as the bottleneck's outflow depends on both how merging flows behave and the bottleneck's available capacity. As previously stated, the studies cited [89], [90] employed adaptive dynamic programming to acquire HRI knowledge in real-time and subsequently plan the robot's movements. Nevertheless, adaptive dynamic programming utilizes pedestrian characteristics, such as their positions and velocities, derived from visual environmental data. Extracting these features from the images requires implementing an additional tracking system.

The rich information in the images regarding pedestrians and HRI prompted the authors to directly use raw images for effective HRI control. Therefore, they introduced an end-to-end learning control framework that connects raw image data to robot actions [22]. Robot motion decisions are generated by the end-to-end model, which derives them from image-observed environment-state features, all to ensure optimal pedestrian flow. The proposed control scheme is depicted in Fig. 7. A DRL-based approach was developed for the formulated robot motion planning problem, as shown. The proposed strategy, informed by environmental imagery, identifies the robot's most effective motion to optimize pedestrian outflow. By directly processing image data of pedestrian flows, the paper's proposed method achieves end-to-end motion planning. These bypasses feature detection and extraction, resulting in faster processing and more efficient online learning than [90].

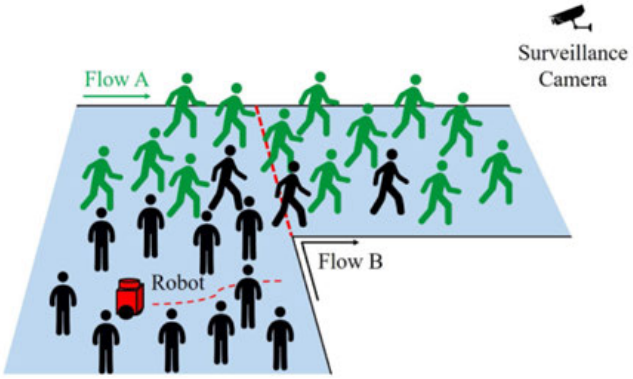


FIGURE 6. The pedestrian flow control scenario defined in [89].

C. ROBOT-ASSISTED PEDESTRIAN GUIDANCE

In today's world, robots are increasingly present in our daily lives, making it crucial to develop ways for them to interact and engage smoothly with people. The integration of robots into human-populated environments is becoming more common, with examples including pedestrian streets, universities, hospitals, and shopping centers. To this point in our discussion, two distinct applications of the robot-assisted pedestrian scheme, specifically encompassing evacuation strategies and flow control mechanisms, have been introduced. This section will focus on the guidance challenge. It is important to note that this category shares some objectives with the two preceding sections and cannot be entirely distinguished from them; however, given its distinct use, we have decided to present the guidance topic in a separate section. Investigations into robots providing human guidance have been carried out in museums [92], at railway stations [93], as supporting aids [94], and in shopping

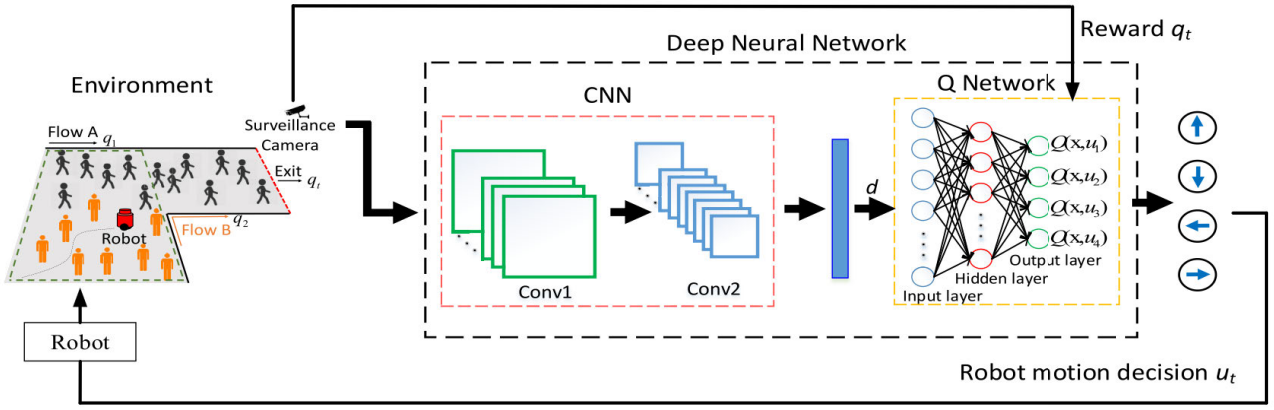


FIGURE 7. The robot-assisted system framework for urban crowd evacuation [84].

centers [95]. This means that the capacity to support, guide, and communicate with people in these evolving contexts will be vital for upcoming robotic systems.

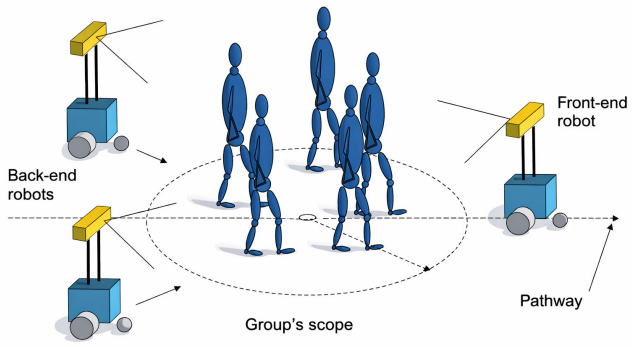


FIGURE 8. Guiding a group of people with a team of mobile robots [96].

The architecture and elements of a motion planner, as described in paper [96], can be used by mobile robots to guide the trajectories of a group of people. This research aims to explore how to provide guidance concerning the difficulty posed by the lack of communication between robots and humans. This system operates analogously to the guidance provided by human professionals in settings like corporate environments, guided tours, and museum exhibits. The control of human trajectories in the suggested scenario involves deploying a fleet of mobile robots to encircle a designated group of individuals, as shown in Figure 8. The lead robot guides, and the trailing robots observe the group to achieve a global perspective and to manage the group's movement if needed. The robot team can perceive their surroundings in real time from multiple viewpoints using stereo vision and share this sensory data with a central system to construct a comprehensive model of the environment, specifically focusing on a group of people. To simulate collective behavior, the social force model was applied in this research. After observing the robots' movements,

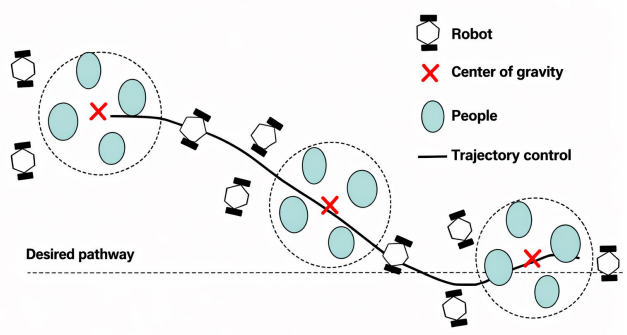


FIGURE 9. The group of people are being led by a team of robots [96].

such as their tendency to cluster according to position and speed, how pedestrians detect approaching robots from behind, the formation of three robots surrounding people, and how the group's center of gravity affects robot motion planning, the model was updated (see Fig. 9). The interplay between human collectives and their robotic counterparts is a key research focus in social and cooperative robotics. Garrell and Sanfeliu [97] developed a model of the environment that enables robots to fulfill their assigned duties. The results of their investigation stemmed from an analysis of how a team of robots collectively assisted individuals in navigating from a designated starting point to a specified destination within an urban setting. The authors herein validate a model previously detailed in [98], termed the "Discrete-Time Motion" (DTM) model. This DTM model presents an innovative method where multiple robots cooperate to guide people in public urban environments. One robot is designated as the leader and functions as a human tour guide. It's positioned at the front of the group, tasked with predicting the paths of both individuals and other robots. The other robots designated as 'shepherds' have the duty of guiding individuals to remain within the group and follow the path set by the leader. The DTM model uses potential functions to represent robots, people, and obstacles, and it estimates the locations of humans and

robots. In a further development of their research, the authors introduced a model for prediction and anticipation (PAM) [99]. This model uses a particle filter (PF) to predict the behavior of a group of people. It also allowed them to identify the optimal robot placement for preventing people from deviating from the formation in specific map zones (anticipation). The PAM was implemented with the specific intention of ensuring individuals remained within the guided group, thereby simplifying the robot's operational duties. Additionally, they could optimize local tasks performed by both robots and humans, resulting in smoother and more human-friendly motion. They ran multiple simulations to verify the model's performance. Due to technological limitations and safety worries, they couldn't fully test the method in real-world situations. This was because there weren't enough high-acceleration robots that were also safe for urban environments. Nevertheless, they successfully assessed the impact of a robot's pushing and dragging forces on individuals in practical scenarios with a single robot. Furthermore, the authors presented an interactive framework for robotic navigation with human awareness, leveraging the Social Force Model and predictive data, as detailed in [100]. They also introduce a novel metric to assess robot companion performance, utilizing vital spaces and comfort as criteria. Additionally, a multimodal human feedback approach is suggested to improve the system's performance. A method for side-by-side social robot navigation, enabling approach and interaction with individuals, is described in [101]. The described methodology enables the robot to escort an individual and approach others, modifying its navigation strategy to anticipate subsequent engagements with people or encounters with stationary obstructions. The act of walking side-by-side is characterized as a collaborative undertaking in which both the human and the robot participants are jointly committed to accomplishing a common objective. This requires the robot to determine the destination and to be able to modify its navigation based on its partner's movements. A significant challenge is the requirement for the robot to effectively predict and learn the person's behavior, anticipate their every movement, and respond in a timely manner. An extended social force model was applied to their simulation, with validation achieved through a comprehensive set of simulations and real-world robot trials.

Despite this, the unpredictable nature of crowds and their decision-making necessitates more flexible, adaptable guidance plans for complex scenarios [102]. While prior research has sought to enhance the practical utility of robot navigation systems through path optimization, improved dynamic obstacle avoidance, and refined human-robot interaction, the study of robots' behavioral influence on crowds of numerous individuals remains underdeveloped [51]. To overcome the limitations of previous research, the study in [51] introduces an innovative, behavior -driven method for human-robot collaborative navigation. It was suggested that navigation robots be configured with different behavioral modes, enabling dynamic modification of their navigation behavior according to real-time data to fulfill the needs of various scenarios by employing a data-driven decision-making mechanism. Moreover, a multi-robot cooperative scheduling algorithm was implemented to address crowd flow management, empowering robots to collectively manage crowd density in designated areas. The research findings facilitate progress in intelligent navigation technology and are crucial for crowd management studies within the context of digital governance.

IV. CONCLUSION AND DISCUSSION

This study attempts to visualize how robotic systems might be incorporated into city life in the coming years, especially in pedestrian control. Scholarly interest in this area is evidently on the rise, as evidenced by the substantial and growing number of publications documented in this investigation. The swift progress in sensor and communication technologies, alongside the creation of equipment that enables robots to perform at progressively higher levels, stands out as a major factor influencing this situation. Urbanization is increasing the frequency and intensity of pedestrian crowding, while cities and transport hubs are becoming denser, more multifunctional, and more dynamic. Under these conditions, conventional crowd-management approaches based on static infrastructure, physical barriers, or signage are often insufficient, especially when the environment changes rapidly or when emergency conditions require an immediate response. Robots offer an attractive alternative because they can function as adaptive, mobile, and intelligent control elements within pedestrian spaces. The reviewed literature indicates that robot-assisted pedestrian control has developed into a promising and highly relevant research direction at the intersection of robotics, transportation, control, and human behavior. The intention was to review the most crucial publications in each category and summarize the progress made to date.

A key insight emerging from the literature is that the role of robots in pedestrian environments extends beyond simple guidance. In many reviewed studies, robots act as dynamic obstacles, local coordinators, mobile information sources, or agents capable of shaping pedestrian movement patterns in real time. This is an important conceptual shift. Traditional infrastructure modifies the environment passively and in a fixed way, whereas robots can sense, interpret, and react. As a result, robot-assisted systems have the potential to improve pedestrian-flow distribution, reduce bottleneck pressure, support safer merging movements, and guide individuals toward safer or more efficient routes. This flexibility is particularly valuable in evacuation scenarios, where conditions can evolve within seconds and where static systems may fail to respond adequately to uncertainty, panic, or blocked paths. At the same time, the near future of robot-assisted pedestrian control will depend on whether current research can move beyond controlled simulations and

simplified laboratory settings. Numerous difficulties persist in translating the insights gained from this research into tangible, real-world applications, stemming from the restrictive nature of experimental parameters and the presence of significant ethical dilemmas. Digital technologies, such as virtual reality, are now being used by researchers to overcome these problems and conduct related research. The study of urban governance and sustainable development has progressively incorporated digital tools as crucial media and innovative methodologies. The capability of digital technology to generate spatial environments nearly indistinguishable from reality enables the simulation of diverse scenarios, equipping researchers with the means to meticulously analyze participants' spatial and social behaviors without compromising safety or ethical guidelines. The existing body of work largely establishes proof of concept; however, achieving genuine deployment hinges upon developing more robust and seamlessly integrated systems for sensing, decision-making, and socially acceptable interaction. For robots deployed in densely populated areas, a critical requirement is their ability to consistently recognize the presence of people, make educated guesses about what those individuals intend to do, pinpoint any hazardous situations, and modify their controlling actions on a continuous, real-time basis. This endeavor is fundamentally dependent upon advancements in the intricate field of sensor fusion, alongside breakthroughs in onboard artificial intelligence, the implementation of efficient distributed communication systems, and a substantial increase in computational efficiency. Equally important is the need for models that capture not only pedestrian motion but also human reactions to robots. Addressing the lack of a unified model designed to illustrate the dynamics of interaction within this mixed traffic context is a crucial area that the current body of research has yet to thoroughly explore. A robot that is optimal from a control perspective may still fail if pedestrians do not trust it, misunderstand its intentions, or respond in unexpected ways. For this reason, human-robot interaction will likely become one of the decisive themes in the next stage of research. Future systems will need to be socially aware, not only collision-free. This means that robots should communicate clearly, move predictably, respect social norms, and adapt to local behavioral patterns. In addition, the increasing reliance on connected and intelligent robotic systems introduces cybersecurity risks that cannot be overlooked. Robot-assisted crowd management systems may be vulnerable to malicious interference, data manipulation, or communication disruptions, which could compromise system reliability and safety. Unauthorized access or adversarial control actions could potentially misguide pedestrians or destabilize crowd flow in critical situations. Therefore, ensuring secure communication, resilient control architectures, and robust protection against cyber threats is essential for safe deployment.

The emergence of more human-like interfaces, including speech, gestures, visual signals, and humanoid embodiments, may improve compliance and emotional acceptance, especially in stressful situations. However, such developments also raise new questions. Human-like robots may increase trust, but they may also create unrealistic expectations about robot capabilities. Consequently, forthcoming efforts should prioritize not solely technological advancement but also the meticulous alignment of user confidence, transparent communication, and the psychological ramifications of robotic integration within public spaces. Its practical success will likely depend on closed-loop intelligent systems that combine robots with cameras, infrastructure sensors, wireless communication, digital maps, and possibly smart signage. In such an architecture, robots become one component of a broader crowd-management ecosystem. This integrated view is especially important for smart-city applications, where robots could cooperate with transportation infrastructure, station management systems, emergency services, and digital twins of urban space. Soon, such integration may allow robots to support everyday pedestrian operations during peak demand while also serving as first-response agents during incidents, disruptions, or evacuations. Overall, the literature supports the conclusion that robot-assisted pedestrian control is no longer a speculative idea but an emerging practical direction with strong potential. Robots cannot fully replace infrastructure, human staff, or established evacuation planning procedures, but they can complement them in meaningful ways. Their main advantage lies in adaptability: they can reconfigure their behavior to match changing crowd conditions, provide localized and responsive guidance, and intervene in situations where static systems are limited. In this sense, the future indeed may "walk with us," but successful realization will require more than technological enthusiasm. It will require interdisciplinary research, realistic experiments, human-centered design, and careful validation under operational conditions. In conclusion, robot-assisted pedestrian management offers a significant opportunity to improve safety, efficiency, and resilience in urban crowds. In the upcoming period, a primary objective should involve the successful transition of innovative ideas from theoretical simulations into practical, real-world testing environments, while simultaneously enhancing robots' abilities to navigate and interact in socially intelligent ways, and furthermore, integrating these robotic systems into comprehensive smart-city control infrastructures. If these challenges are addressed successfully, robots may become valuable partners in pedestrian guidance, flow regulation, and emergency response.

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